

Probing Accretion in M33 X-8 through Broadband X-ray Spectroscopy

Technical Report

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Abstract

M33 X-8 is the nearest ultraluminous X-ray source (ULX) and has been interpreted as a black hole accreting close to the Eddington rate in a very high spectral state. A broadband (0.3–20 keV) X-ray spectral analysis of this source is presented, using two NuSTAR observations and a non-simultaneous Swift/XRT observation, as an independent reproduction of the analysis of Krivonos et al. (2018). Six phenomenological and four physically motivated models are compared. Single-component models fit poorly ($\chi^2/\text{d.o.f.} \geq 1.4$), whereas models combining a thermal disc with a curved or Comptonised high-energy component reach $\chi^2/\text{d.o.f.} \approx 1.1$. In these models the inner disc temperature is approximately 1 keV, in agreement with previous work. However, the electron temperature in the NTHCOMP models is only weakly constrained (13 ± 26 and 17.5 ± 57 keV), whereas Krivonos et al. obtained a tight constraint for one of the same models, and the fit statistics obtained are systematically higher than theirs. Differences in the Swift/XRT spectral extraction are discussed as a possible contributor. An extension of the analysis to XMM-Newton data is in progress.

1 Introduction

Ultraluminous X-ray sources (ULXs) are off-nuclear X-ray sources with luminosities above $\sim 10^{39}$ erg s $^{-1}$, which exceeds the Eddington luminosity of a typical stellar-mass black hole [4]. They are interpreted either as compact objects accreting above the Eddington rate or as accreting black holes of higher mass [4]. Broadband X-ray spectra are a key diagnostic of the accretion flow: the soft emission traces the accretion disc, while the emission above ~ 10 keV probes a hot Comptonising medium, and the relative strength and shape of these components indicate the accretion state.

M33 X-8 is the bright nuclear X-ray source of the galaxy M33, at a distance of about 832 kpc, and is one of the closest known ULXs. Its 0.3–10 keV luminosity of $\sim 1.4 \times 10^{39}$ erg s $^{-1}$ lies near the boundary of the ULX class, which makes it an important source for connecting Galactic X-ray binaries with the more luminous ULXs [2].

Two studies have examined its broadband spectrum. Krivonos et al. [1] analysed two NuSTAR

observations from 2017 together with Swift/XRT data and found a standard disc with an inner temperature of about 1 keV plus a power law with $\Gamma \approx 3$, detected up to 20 keV, with an indication of a high-energy cutoff corresponding to a Comptonising-medium temperature above ~ 10 keV. They concluded that M33 X-8 resembles black hole X-ray binaries in a very high state, accreting close to the Eddington rate, in contrast to the super-Eddington accretion believed to operate in more luminous ULXs. West et al. [2], using nearly simultaneous NuSTAR and XMM-Newton observations, found that an additional high-energy (> 10 keV) Comptonisation component is required, ruling out single advection-dominated disc and classical sub-Eddington models.

In this report, the broadband spectral analysis of Krivonos et al. is reproduced using the same NuSTAR and Swift/XRT observations with an independent data reduction. Six phenomenological and four physically motivated models are fitted in order to (i) characterise the disc and high-energy components, (ii) test which spectral parameters are robust to the choice of model, and (iii) investigate the differences between the results obtained and the published ones, in particular the constraint on the electron temperature. The analysis is being extended to XMM-Newton data. Section 2 describes the data reduction and spectral fitting, Section 3 presents the results and the comparison with previous work, and Section 4 summarises the findings and ongoing work.

2 Methodology

2.1 Data Reduction

2.1.1 NuSTAR.

Two NuSTAR observations of M33 X-8, ObsIDs 50310002001 and 50310002003, obtained in 2017 March and July, were analysed. The data were processed using the NuSTAR Data Analysis Software (NuSTARDAS) within HEASoft. Source and background spectra were extracted using circular and annular regions, respectively, following the extraction geometry adopted in Krivonos et al. [1]: a $70''$ source region and a $100''$ – $160''$ background annulus. The FPMA and FPMB spectra were fitted jointly with a multiplicative constant to account for cross-calibration differences.

2.1.2 Swift/XRT.

The Swift/XRT observation ObsID 00031042002, obtained in photon-counting mode on 2012 November 5, was used. The spectrum was reduced independently using `xselect` and `xrtproducts`, rather than the online extraction procedure used by Krivonos et al. [1]. The resulting spectrum was used to provide soft X-ray coverage below the NuSTAR energy range.

2.2 Spectral Fitting

The spectra were grouped to a minimum of 30 counts per bin and fitted jointly in XSPEC using χ^2 statistics. The NuSTAR and Swift/XRT spectra were fitted over the 3–20 keV and 0.3–5 keV energy ranges, respectively. A multiplicative constant was included to account for cross-calibration differences and non-simultaneity between the observations. Parameter uncertainties are quoted at the 90% confidence level.

2.3 Spectral Modelling

All models include photoelectric absorption and a cross-calibration constant. Two classes of models were considered.

Phenomenological models. Absorbed Power Law (PL), Cutoff Power Law (CPL), DISKBB, and DISKPBB models were considered first. Two-component models, DISKBB+PL and DISKBB+CPL, were then used to describe the spectrum with separate thermal and high-energy components.

Physically motivated models. Comptonisation models in which disc photons provide the seed population for the high-energy component were also considered. These include SIMPL $\otimes$ DISKPBB, SIMPL $\otimes$ DISKPBB $\times$ SPEXPCUT, DISKPBB+NTHCOMP, and DISKBB+NTHCOMP. For the NTHCOMP models, the seed-photon temperature was linked to the inner disc temperature.

In the physically motivated models, the hydrogen column density was fixed at the Galactic value of $N_{\text{H}} = 0.110 \times 10^{22} \text{ cm}^{-2}$, following Krivonos et al. [1].

3 Results

3.1 Phenomenological Models

The best-fit parameters of the six phenomenological models are given in Tables 1–3, and the corresponding spectral fits and residuals are shown in Figure 1. Uncertainties are quoted at the 90% confidence level.

The absorbed Power Law model provides a first description of the broadband continuum and gives $\chi^2/\text{d.o.f.} = 1991/930$, with a steep photon index of $\Gamma = 3.229 \pm 0.012$. However, the residuals show systematic deviations from the model, particularly towards higher energies, indicating that a simple power-law continuum does not capture the observed spectral shape. Introducing an exponential cutoff gives a substantial improvement, reducing χ^2 by 676 for one additional free parameter and yielding $\chi^2/\text{d.o.f.} = 1315/929$. The fitted cutoff energy, $E_{\text{fold}} = 4.27 \pm 0.16 \text{ keV}$, indicates that the hard component is strongly curved rather than extending as an unbroken power law. A similar rollover at approximately 5 keV was reported by Krivonos et al. [1] for

the broadband spectrum of M33 X-8.

A p-free disc model, DISKPBB, was also tested to examine whether the broadband spectrum could be described by a single disc with a modified radial temperature profile. The fit gives $\chi^2/\text{d.o.f.} = 1629/929$, with $kT_{\text{in}} = 1.974 \pm 0.024$ keV. The fitted temperature-profile index reaches $p = 0.500$, corresponding to the lower boundary of the model range. This behaviour is similar to that found by Krivonos et al. [1], who also obtained $p = 0.5$ at the hard limit. In that analysis, however, the single DISKPBB model left a strong excess above ~ 10 keV, and the broadband spectrum was found not to require a slim-disc interpretation. The residuals in the present analysis likewise indicate that a single disc component does not account for the full broadband spectrum.

The next step is therefore to model the spectrum as a combination of a thermal disc and a separate high-energy component. The DISKBB+Power Law model gives $\chi^2/\text{d.o.f.} = 1079/928$, with $kT_{\text{in}} = 1.269 \pm 0.013$ keV and $\Gamma = 2.477 \pm 0.051$. The improvement relative to the single-component models indicates that the broadband spectrum is better represented by distinct soft and hard components. This is consistent with the interpretation of Krivonos et al. [1], who found a disc component with a temperature of approximately 1 keV together with a high-energy tail extending to 20 keV.

Since the single-component cutoff power law indicates significant high-energy curvature, the high-energy component was further replaced by a Cutoff Power Law. The resulting DISKBB+Cutoff Power Law model gives $\chi^2/\text{d.o.f.} = 1009/927$, a further reduction of $\Delta\chi^2 = 70$ for one additional free parameter. The disc temperature is $kT_{\text{in}} = 1.081 \pm 0.022$ keV and the cutoff energy is $E_{\text{fold}} = 5.44 \pm 0.98$ keV. The photon index is $\Gamma = 1.063 \pm 0.373$, although it is relatively poorly constrained. Krivonos et al. [1] obtained a similar spectral shape for the corresponding model, with $kT_{\text{in}} = 1.05 \pm 0.03$ keV and $E_{\text{fold}} = 5.3 \pm 0.4$ keV.

Taken together, the phenomenological fits indicate that the broadband spectrum is not adequately represented by a single continuum component. The spectral shape is better described by a thermal disc with $kT_{\text{in}} \sim 1$ keV together with a distinct high-energy component that exhibits curvature at several keV. These results motivate the physically motivated Comptonisation models considered in the following section.

Table 1: Best-fit parameters for the Power Law and Cutoff Power Law models applied to the broadband X-ray spectrum of M33 X-8.

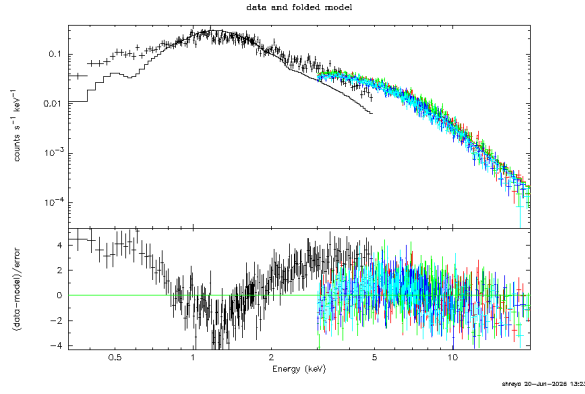
Parameter	Power Law	Cutoff Power Law
N_{H} (10^{22} cm $^{-2}$)	0.656 ± 0.012	0.312 ± 0.013
Constant	1.69 ± 0.04	1.10 ± 0.03
Γ	3.229 ± 0.012	1.762 ± 0.053
E_{fold} (keV)	–	4.27 ± 0.16
$N_{\text{pow}} / N_{\text{cut}}$ at 1 keV	$(1.37 \pm 0.03) \times 10^{-2}$	$(7.32 \pm 0.21) \times 10^{-3}$
$\chi^2/\text{d.o.f.}$	1991/930	1315/929

Table 2: Best-fit parameters for the DISKPBB and DISKBB models applied to the broadband X-ray spectrum of M33 X-8.

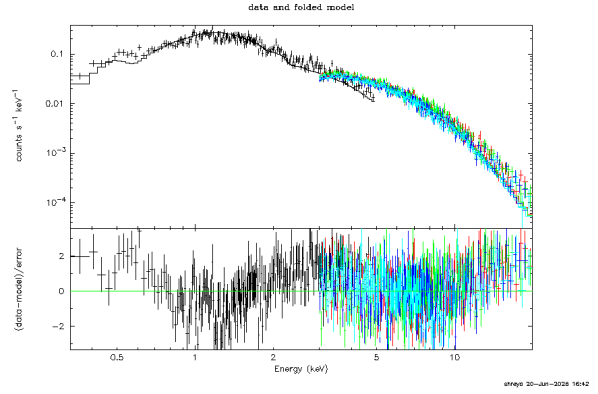
Parameter	DISKPBB	DISKBB
N_{H} (10^{22} cm $^{-2}$)	0.296 ± 0.013	0.0478 ± 0.0053
Constant	0.982 ± 0.027	0.755
kT_{in} (keV)	1.974 ± 0.024	1.575 ± 0.008
p	0.500 (at lower limit)	–
N_{disk}	$(1.34 \pm 0.15) \times 10^{-2}$	0.152 ± 0.003
$\chi^2/\text{d.o.f.}$	1629/929	2299/930

Table 3: Best-fit parameters for the DISKBB+Power Law and DISKBB+Cutoff Power Law models applied to the broadband X-ray spectrum of M33 X-8.

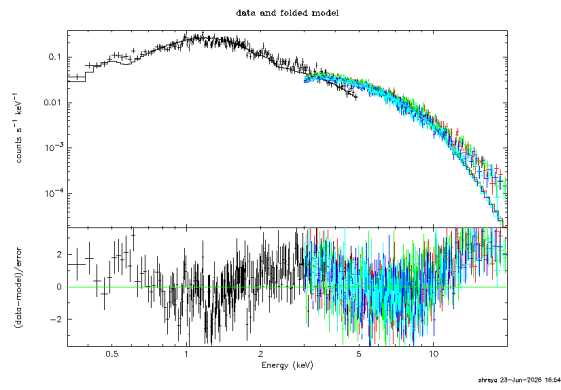
Parameter	DISKBB + Power Law	DISKBB + Cutoff Power Law
N_{H} (10^{22} cm $^{-2}$)	0.264 ± 0.019	0.114 ± 0.023
Constant	0.953 ± 0.024	0.940 ± 0.024
kT_{in} (keV)	1.269 ± 0.013	1.081 ± 0.022
N_{disk}	0.212 ± 0.013	0.451 ± 0.063
Γ	2.477 ± 0.051	1.063 ± 0.373
E_{fold} (keV)	—	5.44 ± 0.98
$N_{\text{pow}} / N_{\text{cut}}$ at 1 keV	$(3.00 \pm 0.34) \times 10^{-3}$	$(0.956 \pm 0.515) \times 10^{-3}$
$\chi^2/\text{d.o.f.}$	1079/928	1009/927



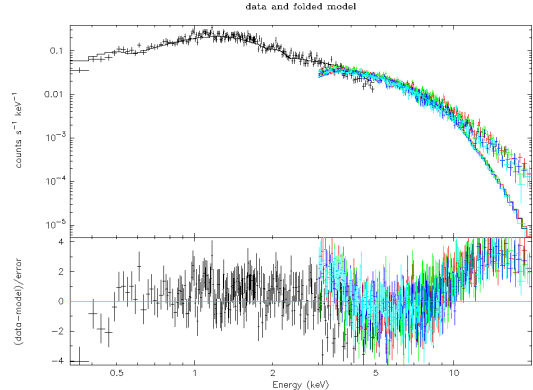
(a) Power Law



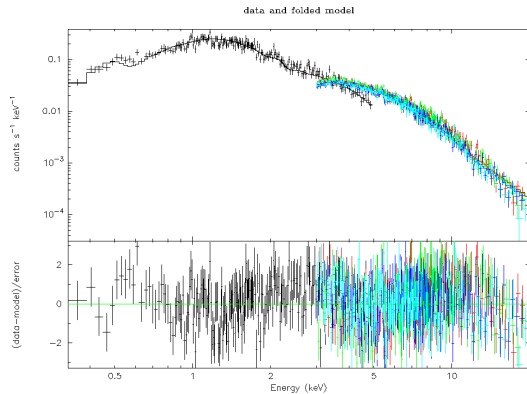
(b) Cutoff Power Law



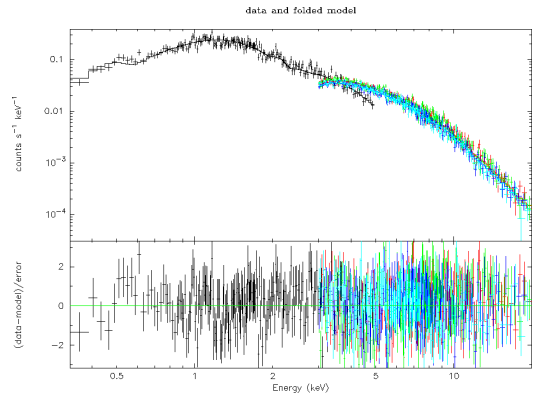
(c) DISKPBB



(d) DISKBB



(e) DISKBB + Power Law



(f) DISKBB + Cutoff Power Law

Figure 1: Phenomenological spectral fits to the broadband X-ray spectrum of M33 X-8. For each model, the upper panel shows the observed spectra together with the folded best-fit model, while the lower panel shows the corresponding residuals, $(\text{data} - \text{model})/\text{error}$.

3.2 Physically Motivated Models

The best-fit parameters of the four physically motivated models are given in Tables 4 and 5. The spectral fits and residuals are shown in Figure 2, and the unfolded spectra are shown in Figure 3. In all four models, N_{H} is frozen at $0.110 \times 10^{22} \text{ cm}^{-2}$, corresponding to the Galactic

absorption adopted for M33 X-8 [1].

The physically motivated models provide comparable descriptions of the broadband spectrum, with $\chi^2/\text{d.o.f.}$ ranging from 1021/928 for SIMPL $\otimes$ DISKPBB to 1005/928 for DISKBB+NTHCOMP. The two NTHCOMP-based models give the lowest χ^2 , while the two SIMPL-based models give very similar fit statistics.

For SIMPL $\otimes$ DISKPBB, the fitted disc temperature is $kT_{\text{in}} = 0.909 \pm 0.053$ keV, with a temperature-profile index of $p = 0.763 \pm 0.024$, close to the value expected for a standard disc, and a scattered fraction of $f_{\text{scat}} = 0.506 \pm 0.076$. Adding SPEXPCUT gives a similar fit, with $\chi^2/\text{d.o.f.} = 1012/927$, and gives $E_{\text{cut}} = 7.15 \pm 1.31$ keV, $kT_{\text{in}} = 1.172 \pm 0.067$ keV, $p = 0.749 \pm 0.018$, and $f_{\text{scat}} = 0.412 \pm 0.045$. The photon index changes from $\Gamma = 3.459 \pm 0.089$ without the cutoff to $\Gamma = 1.738 \pm 0.304$ when the cutoff is included, reflecting the strong dependence of the high-energy continuum parameters on the treatment of the spectral rollover.

The two NTHCOMP-based models give the lowest χ^2 , with $\chi^2 = 1005$ for both models. DISKPBB+NTHCOMP gives $\chi^2/\text{d.o.f.} = 1005/927$, with $kT_{\text{in}} = 0.971 \pm 0.069$ keV and $p = 0.760 \pm 0.080$, while DISKBB+NTHCOMP gives $\chi^2/\text{d.o.f.} = 1005/928$, with $kT_{\text{in}} = 0.980 \pm 0.031$ keV. The similar fit statistics indicate that allowing the disc temperature profile to vary does not produce a significant improvement once the Comptonised component is included.

The electron temperatures obtained from the NTHCOMP fits are $kT_e = 17.53 \pm 57.31$ keV for DISKPBB+NTHCOMP and 13.02 ± 26.10 keV for DISKBB+NTHCOMP. In both cases, the large uncertainties indicate that kT_e is only weakly constrained by the broadband data. To examine this further, **steppar** was used to scan the electron temperature for both models. The resulting confidence scans show that the allowed range of kT_e is broad, confirming that the large formal uncertainties are a consequence of weak parameter constraints rather than a well-defined measurement of the electron temperature. This is particularly relevant for the DISKBB+NTHCOMP model, for which the electron temperature was more tightly constrained in the analysis of Krivonos et al. [1].

The unfolded spectra in Figure 3 show the contributions of the thermal disc and Comptonised components to the broadband emission. The parameter confidence plots in Figure 4 further illustrate the constraints and correlations between the fitted NTHCOMP parameters.

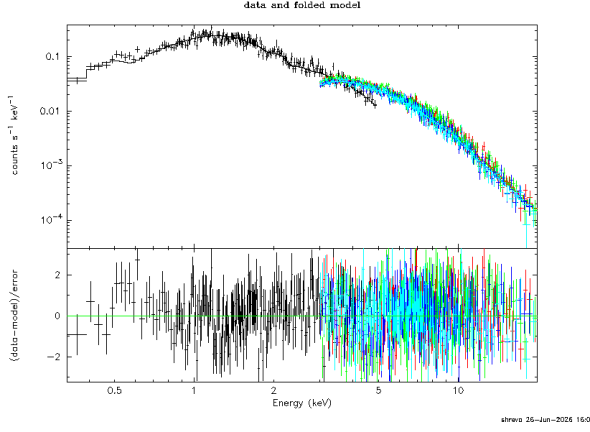
Overall, the physically motivated models give disc temperatures of approximately 0.9–1.2 keV, with $p \approx 0.75$ in the models using DISKPBB. Their similar fit statistics indicate that the present data do not strongly distinguish between the different Comptonisation prescriptions. The main difference between the present analysis and the corresponding results of Krivonos et al. is therefore not the overall spectral shape, but the degree to which the electron temperature is constrained.

Table 4: Best-fit parameters for the $\text{CONST} \times (\text{SIMPL} \otimes \text{DISKPBB})$ and $\text{CONST} \times (\text{SIMPL} \otimes \text{DISKPBB}) \times \text{SPEXPCUT}$ models applied to the broadband X-ray spectrum of M33 X-8.

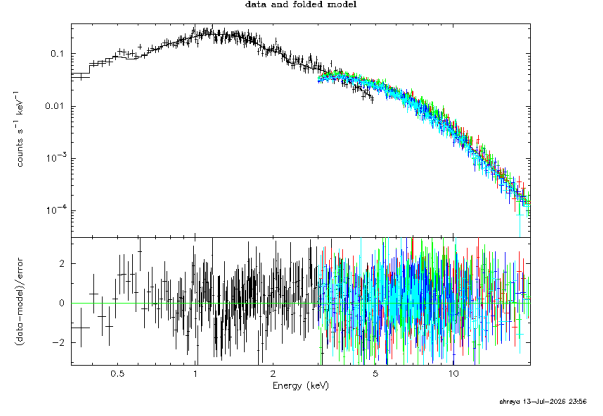
Parameter	$\text{SIMPL} \otimes \text{DISKPBB}$	$(\text{SIMPL} \otimes \text{DISKPBB}) \times \text{SPEXPCUT}$
N_{H} (10^{22} cm^{-2})	0.110 (frozen)	0.110 (frozen)
Constant	1.09 ± 0.04	1.27 ± 0.13
Γ	3.459 ± 0.089	1.738 ± 0.304
f_{scat}	0.506 ± 0.076	0.412 ± 0.045
kT_{in} (keV)	0.909 ± 0.053	1.172 ± 0.067
p	0.763 ± 0.024	0.749 ± 0.018
N_{disk}	1.093 ± 0.300	0.588 ± 0.128
E_{cut} (keV)	–	7.15 ± 1.31
α	–	1.0 (frozen)
$\chi^2/\text{d.o.f.}$	1021/928	1012/927

Table 5: Best-fit parameters for the $\text{CONST} \times (\text{DISKPBB} + \text{NTHCOMP})$ and $\text{CONST} \times (\text{DISKBB} + \text{NTHCOMP})$ models applied to the broadband X-ray spectrum of M33 X-8.

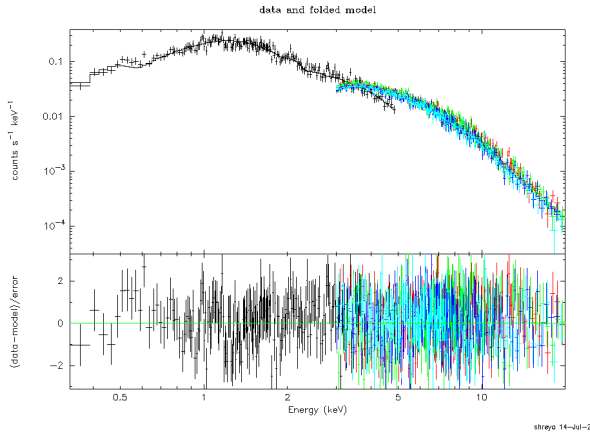
Parameter	$\text{DISKPBB} + \text{NTHCOMP}$	$\text{DISKBB} + \text{NTHCOMP}$
N_{H} (10^{22} cm^{-2})	0.110 (frozen)	0.110 (frozen)
Constant	0.95 ± 0.03	0.95 ± 0.02
kT_{in} (keV)	0.971 ± 0.069	0.980 ± 0.031
p	0.760 ± 0.080	–
N_{disk}	0.334 ± 0.122	0.341 ± 0.109
Γ	3.240 ± 0.682	3.170 ± 0.485
kT_e (keV)	17.53 ± 57.31	13.02 ± 26.10
kT_{bb} (keV)	0.971 (linked)	0.980 (linked)
N_{nth} (10^{-3})	1.93 ± 0.96	1.82 ± 0.53
$\chi^2/\text{d.o.f.}$	1005/927	1005/928



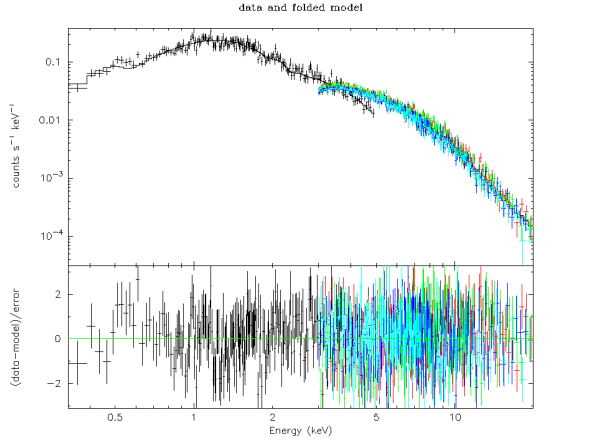
(a) $\text{SIMPL} \otimes \text{DISKPBB}$



(b) $(\text{SIMPL} \otimes \text{DISKPBB}) \times \text{SPEXPCUT}$

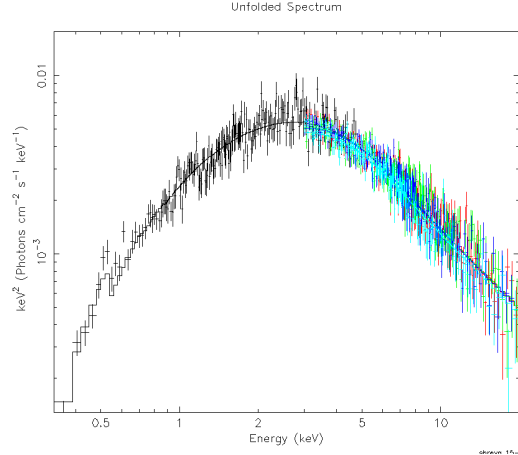


(c) $\text{DISKPBB} + \text{NTHCOMP}$

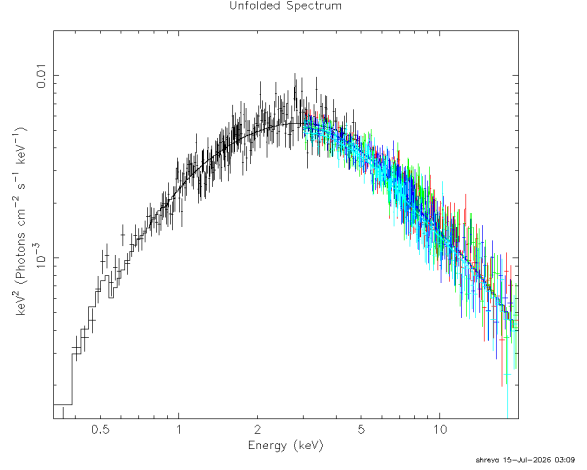


(d) $\text{DISKBB} + \text{NTHCOMP}$

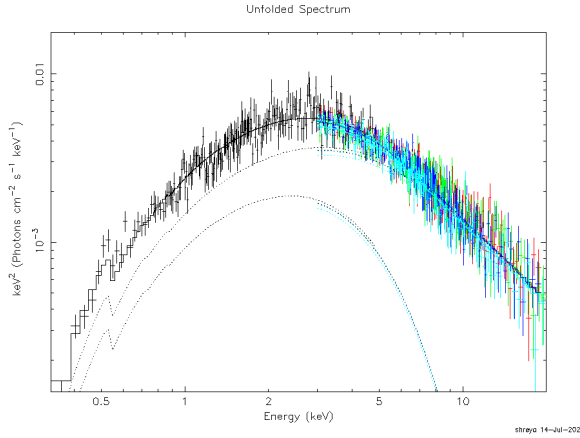
Figure 2: Physically motivated spectral fits to the broadband X-ray spectrum of M33 X-8. For each model, the upper panel shows the observed spectra together with the folded best-fit model, while the lower panel shows the corresponding residuals, $(\text{data} - \text{model})/\text{error}$. The models shown are $\text{SIMPL} \otimes \text{DISKPBB}$, $(\text{SIMPL} \otimes \text{DISKPBB}) \times \text{SPEXPCUT}$, $\text{DISKPBB} + \text{NTHCOMP}$, and $\text{DISKBB} + \text{NTHCOMP}$.



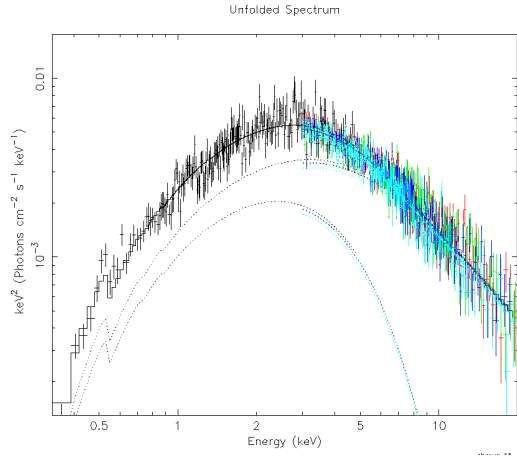
(a) SIMPL $\otimes$ DISKPBB



(b) (SIMPL $\otimes$ DISKPBB)×SPEXPCUT

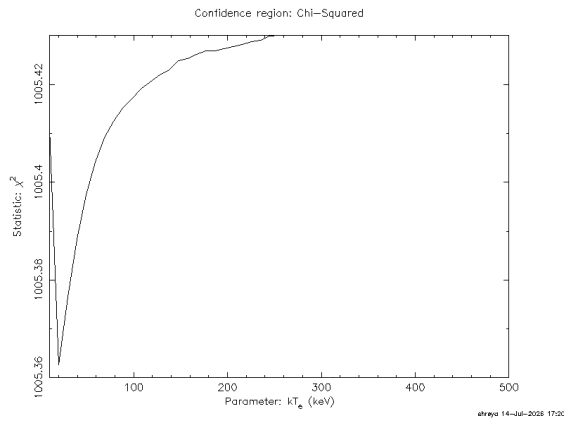


(c) DISKPBB+NTHCOMP

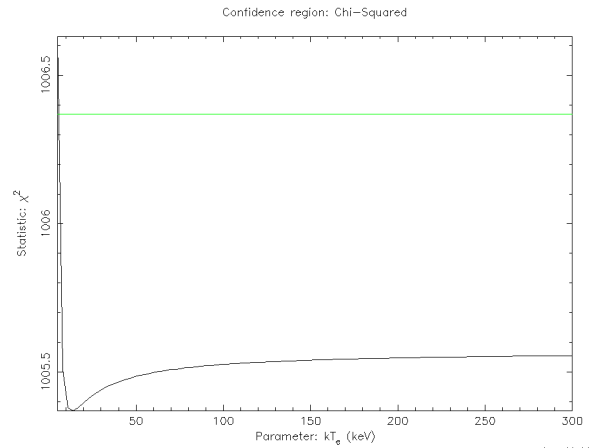


(d) DISKBB+NTHCOMP

Figure 3: Unfolded EF_E spectra for the physically motivated models considered for M33 X-8. The individual thermal disc and Comptonised components are shown where applicable.



(a) DISKPBB+NTHCOMP



(b) DISKBB+NTHCOMP

Figure 4: Confidence contours for the fitted NTHCOMP parameters in the DISKPBB+NTHCOMP and DISKBB+NTHCOMP models. The contours illustrate the parameter correlations and the weak constraint on kT_e .

3.3 Model Comparison

Across the ten models considered, the single-component models provide substantially poorer fits to the broadband spectrum than the two-component models. Among the phenomenological models, the addition of a second spectral component significantly improves the fit, with DISKBB+Cutoff Power Law giving $\chi^2/\text{d.o.f.} = 1009/927$.

The physically motivated models provide comparable fit statistics, ranging from $\chi^2/\text{d.o.f.} = 1021/928$ for SIMPL $\otimes$ DISKPBB to 1005/928 for DISKBB+NTHCOMP. The small differences in fit statistic among the best-fitting models do not provide strong statistical evidence for preferring one Comptonisation prescription over another. However, these models provide a physically motivated description of the spectrum in terms of thermal disc emission and a high-energy Comptonised component.

For the subsequent physical interpretation, DISKBB+NTHCOMP is adopted as the reference model because it provides one of the best fits while allowing the spectrum to be described in terms of physically interpretable disc and Comptonisation parameters. The other physically motivated models are retained to assess the robustness of the inferred spectral properties.

3.4 Comparison with Previous Work

Krivonos et al. [1] found that the broadband spectrum of M33 X-8 is well described by a thermal disc with an inner temperature of approximately 1 keV together with a steep high-energy component, with $\Gamma \approx 3$. Their analysis also found evidence for curvature in the high-energy spectrum and interpreted this in terms of Comptonisation. The results obtained here are broadly consistent with this spectral picture. The physically motivated models give disc temperatures of $kT_{\text{in}} = 0.9\text{--}1.2$ keV, with photon indices of $\Gamma \approx 3.2\text{--}3.5$ in all models except the cutoff SIMPL model, which gives $\Gamma = 1.738 \pm 0.304$. The phenomenological models similarly favour a disc component with a temperature close to 1 keV and a curved high-energy component.

West et al. [2], using nearly simultaneous NuSTAR and XMM-Newton observations of M33 X-8, found that an additional high-energy (> 10 keV) Comptonisation component is required, ruling out single advection-dominated disc and classical sub-Eddington models.

The main discrepancy between the two analyses is the constraint on the electron temperature in the NTHCOMP models. In the present analysis, DISKBB+NTHCOMP gives $kT_e = 13.02 \pm 26.10$ keV, while DISKPBB+NTHCOMP gives $kT_e = 17.53 \pm 57.31$ keV. The uncertainties are large compared with the corresponding best-fit values, indicating that kT_e is not tightly constrained. This differs particularly from the DISKBB+NTHCOMP result of Krivonos et al., who obtained $kT_e = 17.7^{+1.3}_{-2.4}$ keV. For their DISKPBB+NTHCOMP model, however, the spectrum was insensitive to the high-energy rollover and only a lower limit of $kT_e > 100$ keV was obtained.

To investigate whether the large uncertainties in the present NTHCOMP fits reflected a numerical fitting issue or a genuine lack of constraint, **steppar** scans were performed over kT_e for

both NTHCOMP models. The scans confirmed that a broad range of electron temperatures is allowed by the data, supporting the conclusion that kT_e is intrinsically weakly constrained in the present fits. This is particularly important for the DISKBB+NTHCOMP model, for which the published analysis gives a relatively precise measurement of kT_e , whereas the present reduction does not.

Table 6 compares the fit statistics obtained in the present analysis with those reported by Krivonos et al. [1] for the models common to both analyses. The fit statistics are systematically higher in the present analysis, although the relative behaviour of the different models is similar. In particular, the two-component models provide substantially better fits than the corresponding single-component models in both analyses.

A likely source of the difference in the NTHCOMP constraints is the treatment of the Swift/XRT spectrum. Krivonos et al. extracted the Swift/XRT spectrum using the online tools of the UK Swift Science Data Centre. These tools account for pile-up by identifying affected intervals and excluding the piled-up inner part of the point-spread function from the spectral extraction when necessary [1]. In the present analysis, the Swift/XRT spectrum was extracted manually using `xselect` and `xrtproducts`. Thus, although the same Swift/XRT observation is used, the resulting soft X-ray spectrum is not produced by the same extraction procedure.

This difference is important because the electron temperature in NTHCOMP is constrained through the high-energy curvature of the Comptonised component, and its fitted value can be correlated with the continuum parameters that determine the broadband spectral shape. Differences in the extracted Swift/XRT spectrum, particularly in the softer energy range, can therefore affect the balance between the disc and Comptonised components and consequently the inferred constraint on kT_e . The agreement in the overall spectral shape together with the different degree of constraint on kT_e suggests that the extraction procedure is a plausible contributor to the discrepancy, although it cannot be established as the sole cause from the present analysis.

The analysis is therefore being extended to XMM-Newton data. The broader and better-characterised soft X-ray coverage provided by XMM-Newton will allow the soft spectral component to be tested independently and will help determine whether the difference in the NTHCOMP electron-temperature constraint is associated with the Swift/XRT reduction.

Table 6: Comparison of $\chi^2/\text{d.o.f.}$ between this work and Krivonos et al. [1] for the models in common.

Model	This work	Krivonos et al.
Power law	1991/930	1690/903
Cutoff power law	1315/929	1207/902
DISKPBB	1629/929	1584/902
DISKBB + Power law	1079/928	951/901
DISKBB + Cutoff power law	1009/927	891/900
SIMPL $\otimes$ DISKPBB	1021/928	901/901
(SIMPL $\otimes$ DISKPBB) $\times$ SPEXPCUT	1012/927	891/900
DISKPBB + NTHCOMP	1005/927	886/900
DISKBB + NTHCOMP	1005/928	887/901

4 Summary and Ongoing Work

The single-component models cannot adequately describe the broadband spectrum of M33 X-8. A soft disc with $kT_{\text{in}} \approx 1$ keV together with a hard Comptonised or cutoff component provides a substantially better description of the data, consistent with the spectral properties expected for a source in a very high state. The electron temperature, however, is not tightly constrained by the present data. The analysis is being extended to XMM-Newton data, which will provide improved soft-band coverage and help assess the differences between the present results and previous studies.

References

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